Examining the Utility-based Resiliency of Utah's Highway Network

Gregory Macfarlane[a], Max Barnes[a], Natalie Gray[a]

[a] *Brigham Young University Civil and Environmental Engineering Department 430 Engineering Building Provo Utah 84602*

Abstract

This is where the abstract should go.

Keywords: Accessibility, Passive Data, Location Choice

1. Introduction

Systemic resiliency is an important consideration for transportation agencies, though specific definitions of "resiliency" might vary under different contexts. Some agencies and researchers see resiliency as facility-level design and engineering that hardens the system against failure (Bradley, 2007; Peeta et al., 2010); others as an ability for maintenance staff to rapidly restore service following catastrophe (Zhang and Wang, 2016); and others as an ability for a system to continue operating in degraded state (Berdica, 2002; Ip and Wang, 2011). Regardless of the definition used, assessing the resiliency of a transportation network — and addressing any potential shortfalls — requires a method to identify which links or facilities are most critical to the smooth operation of the network.

Identifying which links in a transportation network are most critical, however, is not trivial. State of the practice techniques typically rely on traffic volumes represented in terms of vehicles, trips, or freight value (AEM, 2017). But these methods ignore the fact that some networks have alternate routes readily available, have multiple well developed modes of transportation, or have redundant destination locations. When a network link does break, routes, mode choice, and destination choice often change too. These responses have been observed in real-world crisis events like the I-35 W bridge collapse in Minneapolis and the I-85 bridge fire and collapse in Atlanta (Zhu et al., 2010b,a; Xie and Levinson, 2011; Hamedi et al., 2018).

In this research, we apply destination choice accessibility model to identify critical highway links in a statewide highway context. This model is designed to capture the utility lost to individuals operating on a degraded network who may choose a longer route, a different travel mode, or a different destination. We then apply the model to a highway network representing the state of Utah. The model structure permits a more nuanced evaluation of link criticality than using traffic volume alone or in conjunction with an increased travel time measure.

The paper proceeds as follows. First, a literature review discusses previous attempts to analyze the resiliency of transportation networks. A methodology section presents the model developed for this research and describes the implementation process in Utah. A results section describes the model output in detail for one scenario and then compares and ranks the model outputs for an array of several dozen independent scenarios. The paper concludes with a discussion of limitations and associated avenues for future research.

*Corresponding author

Email addresses: gregmacfarlane@byu.edu (Gregory Macfarlane), barnesmaxmax@gmail.com (Max Barnes), nat.gray2000@gmail.com (Natalie Gray)

[1] Corresponding Author

Table 1: Attempts to Evaluate Systemic Resiliency

Year	Author	Performance Metric
2004	Geurs and van Wee	Accessibility (isochrone, gravity, logsum)
2007	Abdel-Rahim et al.	Network Connectivity
2008	Taylor, M	Accessibility (logsum)
2010	Peeta et al.	Travel time and cost
2010	Geurs et al.	Accessibility (logsum)
2010	Levinson and Zhu	Travel time and cost
2010	Zhu et al.	Travel time and cost
2011	Agarwal et al.	Network connectivity
2011	Ip and Wang	Network connectivity
2011	Serulle et al.	Travel time and cost
2011	Ibrahim, S	Travel time and cost
2011	Xie and Levinson	Accessibility (isochrone)
2013	Omer et al.	Travel time and cost
2014	Osei-Asamoah and Lownes	Network connectivity
2015	Zhang et al.	Network connectivity
2015	Guze	Network connectivity
2015	Jaller et al.	Travel time and cost
2015	Xu et al.	Network connectivity
2016	Winkler, C.	Accessibility (gravity)
2017	Ganin et al.	Accessibility (gravity)
2019	Vodak et al.	Network connectivity
2019	Hackl and Adey	Network connectivity

2. Literature

In a groundbreaking theoretical article, Berdica (2002) attempted to identify, define and conceptualize network "vulnerability" — the complement of resiliency — by envisioning analyses conducted with several vulnerability performance measures including travel time, delay, congestion, serviceability, and accessibility. She then defined reliability as the level of reduced accessibility due to unfavorable operating conditions on the network. In particular, Berdica identified a need for further research toward developing a framework capable of investigating reliability of transportation networks.

In this section we examine several attempts by numerous researchers to do precisely this using various measures of network performance. It is helpful to categorize this review into three groups based on the overall technique applied in the study:

- *Network connectivity*: How does damage to a network diminish the connectivity between network nodes?
- *Travel time analysis*: How much do shortest path travel times between origins and destinations increase on a damaged network?
- *Accessibility analysis* How easily can the population using the damaged network complete their daily activities?

The following sections discuss relevant studies in each group; Table 1 consolidates these studies by year and labels them with an applicable group.

2.1. Network Connectivity

The purpose of transportation networks is to connect locations to each other; presumably damage to a network would diminish the network's *connectivity*, or the number of paths between node pairs. It may even leave nodes or groups of nodes completely isolated. In studies based on connectivity, researchers typically apply methods and concepts from graph theory.

Abdel-Rahim et al. (2007) developed a multi-layered graph to examine the resiliency of the traffic signal control system in Boise, Idaho. The researchers determined which traffic signals would be isolated by a failure to a particular power substation, and consequentially the percent of travel paths that would experience diminished levels of service. The research highlights the degree to which interrelated infrastructure systems — power, telecommunications, and transportation — depend on each other, though the researchers did not attempt to look at the connective resiliency of the transportation network directly.

Agarwal et al. (2011) present a method to represent a transportation network as a hierarchical graph that can be analyzed more directly for vulnerabilities. The authors acknowledge, however, that a maximal failure consideration where a node is entirely isolated from the network is unlikely in a real-world network with multiple paths of connectivity.

Ip and Wang (2011) address this shortcoming through the concept of *friability*, or the reduction of capacity caused by removing a link or node, in order to determine criticality of individual links. The methodology relies on the ability to determine the weighted sum of the resilience of all nodes based on the weighted average of connectedness with other city nodes in the network. The authors determine that the recovery of transportability between two cities largely depends on redundant links between nodes. The authors also comment that most traffic managers are more concerned with the friability of single links rather than the friability of multiple links or an entire system.

Vodák et al. (2019) develop an approach to identify critical links in a network by searching for the shortest independent loops in the network. The algorithm progressively damages one or more links between iterations to determine if nodes become isolated, or cut off from the network. If a node becomes easily isolated or has a higher likelihood of becoming isolated, then there is a higher degree of vulnerability present in the network. This method can both identify critical links in individual networks, as well as provide a means to quantitatively compare networks.

All of these graph theoretical approaches tend to break down to some degree on large, real-world networks where the number of nodes and links numbers in the tens of thousands, and the degree of connectivity between any arbitrary node pair is high.

2.2. Changes in Travel Time

Highway system network failures — in most imaginable cases — degrade the shortest or least cost path but typically do not eliminate it entirely. The degree to which travel time increases when a particular link is damaged could provide an estimate of the criticality of that link.

Peeta et al. (2010) construct a model to efficiently allocate highway maintenance resources. Each link in the sample network was assigned a specific failure probability based on resource allocation; the model evaluated the increase in travel time resulting from a broken link. A Monte Carlo simulation revealed which allocation plan resulted in the least network degradation, and thus which links were most critical to the network's operations.

Ibrahim et al. (2011) provide an alternative heuristic approach for determining vulnerability of infrastructure by estimating the cost of single link failure based on the increase in shortest path travel time due to increased congestion levels. The authors propose a hybrid heuristic approach that calculates the traditional user-equilibrium assignment for finding the first set of costs, and then fixes those costs for all following iterations to determine the effects of failure on overall travel time of the system. Omer et al. (2013) apply a similar methodology to a real-life intercity highway network. Jaller et al. (2015) extended this methodology with a static user equilibrium traffic loading step to provide an estimate of how the next-shortest path changes when congested.

A primary limitation with increased travel time methodologies is that they ignore the other possible ways a population might adapt its travel to a damaged network. Some people may choose other modes or destinations, and it is possible that some previously occurring trips might be canceled entirely.

2.3. Changes in Accessibility

In a travel modeling context, *accessibility* refers to the ease with which individuals can reach the destinations that matter to them; this is an abstract idea but one that has been quantified in numerous ways. Dong et al. (2006) provide a helpful framework for understanding various quantitative definitions of accessibility that we will simplify here. The most elementary definition of accessibility is whether a destination is within an *isochrone*, or certain distance. This measure is often represented as a count, e.g., the number of jobs reachable from a particular location within thirty minutes travel time by a particular mode. Mathematically,

$$A_i = \sum_j X_j I_{ij}; I_{ij} = \begin{cases} 1 \text{ if } d_{ij} \leq D \\ 0 \text{ if } d_{ij} > D \end{cases} \tag{1}$$

where the accessibility A at point i is the sum of the all the destinations X at other points j. I_{ij} is an indicator function equal to zero if the distance between the points $d_i j$ is less than some asserted threshold (e.g., thirty minutes of travel time). By relaxing the assumption of a binary isochrone and instead using the distance directly, we can derive the so-called gravity model,

$$A_i = \sum_j X_j f(d_{ij}) \tag{2}$$

where the function $f(d_{ij})$ is often a negative exponential with a calibrated impedance coefficient. An extension of the gravity model is to use the logsum term of a multinomial logit destination choice model,

$$A_i = ln \sum_j \beta_d(d_{ij}) + X_j \beta \tag{3}$$

where the parameters β are estimated from choice surveys or calibrated to observed data. The logsum term has numerous benefits outlined by Handy and Niemeier (1997) and Geurs and van Wee (2004); namely, the measure is based in actual choice theory, and can include multiple destination types and travel times by multiple different modes.

Taylor (2008) applied logsum-derived accessibility analysis to evaluate the consequences of a tunnel failure in Adelaide, Australia.

1. LOOK LITERATURE THAT TAYLOR CITES
2. LOOK AT LITERATURE THAT HAS CITED TAYLOR
3. TELL THE STORY OF WHY WHAT WE DID IS WORTHWHILE

BECAUSE THERE IS A GAP! STATEWIDE

3. Methodology

The objective of this study is to evaluate the relative systemic criticality of highway links on a statewide network using a model sensitive to changes in route path, destination choice, and mode choice. We first describe the model, and then its implementation.

3.1. Model Design

The overall model framework is presented in Figure 1, and is designed to capture the utility-based accessibility for a particular origin zones i and trip purpose m. The model begins with a travel time skim procedure, to determine the congested travel time from zone i to zone j by auto as well as the shortest network distance for non motorized modes. The transit travel time skim is fixed, assuming that transit infrastructure would not be affected by changes to the highway network. Throughout this section, lower-cased index variables k belong to a set of all indices described by the corresponding capital letter K

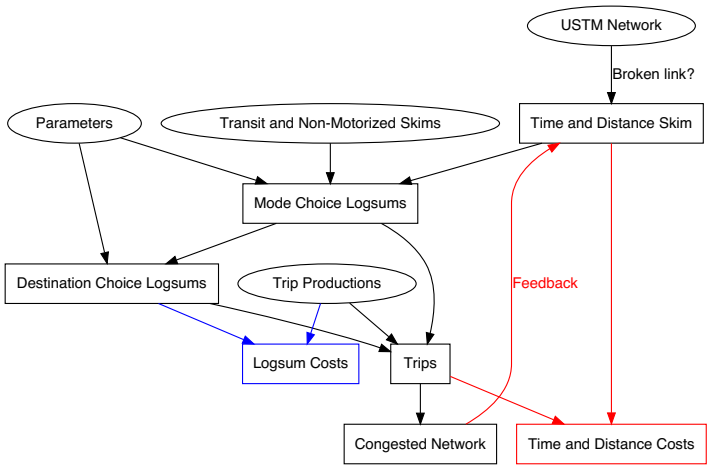


Figure 1: Model framework.

With the travel time t_{ijk} for all modes $k \in K$, the model computes mode choice utility values. The multinomial logit mode choice model describes the probability of a person at origin i choosing mode k for a trip to destination j:

$$\mathcal{P}_{ijm}(k) = \frac{\exp(f(\beta_m, t_{ijk}))}{\sum_K \exp(f(\beta_m, t_{ijk}))} \tag{4}$$

The log of the denominator of the this equation is called the mode choice logsum, $MCLS_{ijm}$ and is a measure of the travel cost by all modes, weighted by utility parameters β_m that may vary by trip purpose.

The $MCLS$ is then used as a travel impedance term in the multinomial logit destination choice model, where the probability of a person at origin i choosing destination $j \in J$ is

$$\mathcal{P}_{im}(j) = \frac{\exp(f(\gamma_m, MCLS_{ijm}, A_j))}{\sum_J \exp(f(\gamma_m, MCLS_{ijm}, A_j))} \tag{5}$$

where A_j is the attractiveness — represented in terms of socioeconomic activity — of zone j. As with mode choice, the log of the denominator of this model is the destination choice logsum, $DCLS_{im}$. This quantity represents the value access to all destinations by all modes of travel, and varies by trip purpose.

The $DCLS_{im}$ measure is relative, but can be compared across scenarios. The difference between the measures of two scenarios

$$\Delta_{im} = DCLS_{im}^{\text{Base}} - DCLS_{im}^{\text{Scenario}} \tag{6}$$

however, provides an estimate of the accessibility lost when $t_{ij\text{drive}}$ changes due to a damaged highway link. This accessibility change is *per trip*, meaning that the total lost accessibility is $P_{im} * \Delta_{im}$ where P is the number of trip productions at zone i for purpose m. This measure is given in units of dimensionless utility, but the mode choice cost coefficient β provides a conversion factor between utility and cost. The total financial cost of a damaged link for the entire region for all trip purposes is

$$\text{Cost} = \sum_I \sum_M -1/\beta_{\text{cost},m} * P_{im}\Delta_{im} \tag{7}$$

For comparison to a simpler resiliency method that only includes the increased travel time between origins and destinations, we compute the change in travel time between δt_{ij} and multiply the number of

trips by this change and a value of time coefficient derived from the cost and vehicle time coefficients of the mode choice model,

$$\text{Cost}' = \sum_I \sum_J \sum_M \frac{\beta_{\text{time},m}}{\beta_{\text{cost},m}} T_{ijm} t_{ijm} \tag{8}$$

3.2. Model Implementation in Utah

The Utah Department of Transportation (UDOT) manages an extensive highway network consisting of interstate freeways (I-15, I-80, I-70, and I-84), intraurban expressways along the Wasatch Front, and rural highways throughout the state. The rugged mountain and canyon topography throughout the state places severe constraints on possible redundant paths in the highway network. A landslide or rock fall in any single canyon may isolate a community or force a redirection of traffic that could be several hours longer than the preferred route; understanding which of these many possible choke points is most critical is a key and ongoing objective of the agency.

Several data elements for the model described above were obtained from the Utah Statewide Travel Model (USTM). USTM is a trip-based statewide model that is focused exclusively on long-distance and rural trips: intraurban trips within existing Metrpolitan Planning Organization (MPO) model regions are pre-loaded onto the USTM highway network. This means that USTM as currently constituted can be used for infrastructure planning purposes, but would be inadequate to evaluate the systemic resiliency of the highway network given the disparate methodologies of the MPO models. USTM can, however, provide the following data elements

1. *Highway Network*: including free flow and congested travel speeds, link length, link capacity estimates, etc.
2. *Zonal Productions* P_{im}: available for all zones by purpose, including those in the MPO region areas.
3. *Zonal Socioeconomic Data*: the destination choice model described in Equation (5) calculates attractions A_{jm} from the USTM zonal socioeconomic data based on the utility coefficients in 2.
4. *Calibration Targets*: USTM base scenario estimates of mode split and trip length were used to calibrate the utility coefficients as described below.

Among MPO models in Utah, only the model jointly operated by the Wasatch Front Regional Council (WFRC, Salt Lake area MPO) and the
Mountainland Association of Governments (MAG, Provo area MPO) model include a substantive transit forecasting component. The transit travel time skim from the WFRC / MAG model was used for the mode choice model in Equation (4); the zonal travel time between the smaller WFRC / MAG model zones was averaged to the larger USTM zones, and the minimum time among the several modes available (commuter rail, light rail, bus rapid transit, local bus) was taken as the travel time for a single transit mode in this implementation.

The utility coefficients for the destination and mode choice models are presented in Table 2. The mode choice coefficients were adapted from USTM and supplemented with coefficients from the Roanoke (Virginia) Valley Transportation Planning Organization (RVTPO) travel model. This model was selected for its simplicity and analogous data elements to the proposed model. The alternative-specific constants were calibrated to regional mode choice targets developed from the 2015 Utah Household Travel Survey (UHTS) using methods described by Koppelman and Bhat (2006). The destination choice utility equation consists of three parts: a size term, a travel impedance term, and a calibration polynomial. Coefficients for the size term and travel impedance terms were adapted from the Oregon Statewide Integrated Model (SWIM) for all purposes except HBW. Instead, these coefficients were adapted from the RVTPO model. The distance polynomial coefficients were calibrated to targets developed from UHTS.

3.2.1. Vulnerable Link Identification

To develop evaluation scenarios on which to apply the model, we used information contained in the UDOT Risk Priority Analysis online map [CITATION]. This map considers the probability of various events that could impact road performance including rock falls, avalanches, landslides, and other similar occurrences. The map indicated 41 non-redundant highway facilities on which to apply the model.

Table 2: Choice Model Coefficients

Variable	HBW	HBO	NHB
Destination Choice			
Households	0.0000	1.0187	0.2077
Office Employment	0.4568	0.4032	0.2816
Other Employment	1.6827	0.4032	0.2816
Retail Employment	0.6087	3.8138	5.1186
Distance	-0.0801	-0.1728	-0.1157
Distance^2	0.0026	0.0034	0.0035
Distance^3	0.0000	0.0000	0.0000
Mode Choice			
Shared	-1.1703	0.0164	-0.0336
Transit	-0.3903	-1.9811	-2.2714
Non-Motorized	-1.2258	-0.3834	-0.8655
Travel Time [minutes]	-0.0450	-0.0350	-0.0400
Travel Cost [dollars]	-0.0016	-0.0016	-0.0016
Walk Distance (less than 1 mile) [miles]	-0.0900	-0.0700	-0.0800
Walk Distance (1 mile or more) [miles]	-0.1350	-0.1050	-0.1200

4. Results

In this chapter, we summarize the methodology used to identify vulnerable links on Utah's highway network. We also apply the model to scenarios where critical highway links are removed from the model network. This includes first a detailed analysis of a single scenario, where I-80 between Salt Lake and Tooele Counties is severed. We compare the model output to an alternative method that measures only the change in travel time and does not allow for mode or destination choice. The model was then applied to 41 individual link closure scenarios throughout the state.

4.1. Vulnerable Link Identification

Two methodologies were developed to identify vulnerable network links in Utah. The first method resembles the process used by AEM to determine threat categories, and threat proximity thresholds to highway links. This methodology was ultimately not used in the resiliency model. The second method uses an online Risk Priority Analysis map created by UDOT combined with familiar knowledge of Utah's road system. Using this second method, links were identified due to their location in relation populations or geography, are suspected choke points, or because they were of interest.

4.2. Single Scenario Analysis

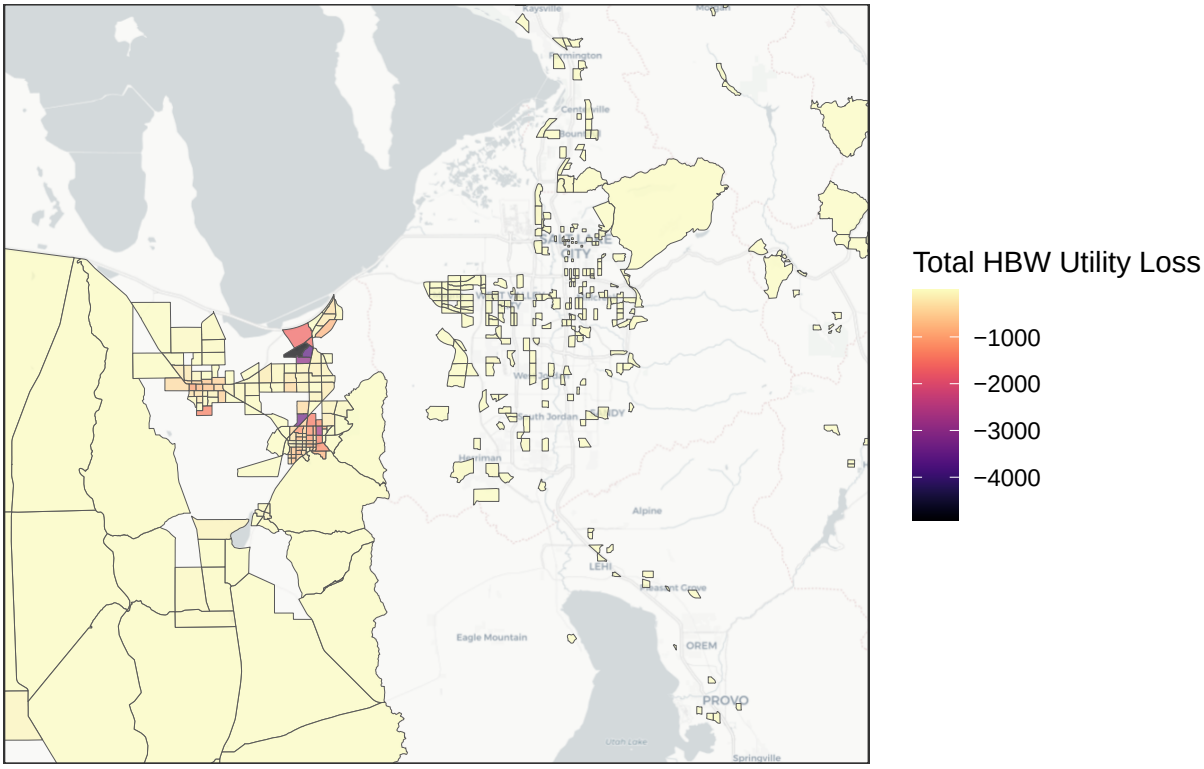
This section outlines an in-depth analysis that was conducted to ensure the resiliency model was accurately capturing trips with OD pairs in the targeted area around a closed link. This analysis was done on a link between Tooele and Salt Lake City, Utah.

Scenario 50, located along I-80 between Tooele and Salt Lake Counties, was examined to ensure the resiliency model was capturing trips in the vicinity of a broken link. Analyzing the model outputs at a localized level was necessary to ensure that the model was appropriately estimating trips and capturing them in the areas that were expected. In scenario 50, if many effected trips had not been captured in Tooele and Salt Lake counties, that would have signaled that a problem was present with the model. This analysis shows that a broken link mainly effects the area surrounding that link, and the key takeaway here is that the model functions as intended. Another method to estimate trip costs is the travel time method, which

serves to capture trips that have more fixed OD pairs, such as freight trips. This analysis also looks at this method in addition to the logsum estimates.

Table **??** compares the overall costs between the logsum and travel time methods, the specific cost for trips originating in Tooele and ending in Salt Lake and includes a trip comparison as well. From Table **??**, we can see that the logsum method captures about $173,996 in experienced expense due to the closure of the link. Specifically, between Tooele and SLC, the logsum based resiliency model captures $167,424 of expense, which is approximately 99% of the total expense as seen in Table **??**. This shows that the resiliency model is effectively capturing trips in the correct areas based on the percent comparison capture rate in Table **??**. When we look at the travel time method of analysis, we can see that the costs at both the local and statewide levels are much greater with $437,401 and $406,899 respectively estimated as the costs due to just the increase in travel time, not using the logit-based model.

Purpose	Other Counties	Inside Tooele	Total	Other Counties	Inside Tooele	Total
HBO	$98,674	$1,193,347	$1,292,020	$8,595	$75,862	$84,457
HBW	$15,140	$380,590	$395,730	$7,868	$234,009	$241,877
NHB	$31,317	$451,263	$482,580	$5,875	$100,245	$106,120
Internal Freight				$24,617	$26,083	$50,700
Inbound / Outbound Freight				$59,758	$1,190	$60,948
REC				$176	$190	$366
Through Freight				$251,508		$251,508
Through Passenger				$56,103		$56,103
Comparable Total	$145,130	$2,025,199	$2,170,330	$22,338	$410,116	$432,454
Total	$145,130	$2,025,199	$2,170,330	$414,499	$437,580	$852,079



The logsum and travel time methods for determining the overall cost associated with can be broken down into overall costs, and comparable costs. The comparable costs are made up of those purposes which are included in the resiliency model. HBW, HBO, and NHB trips are included in both models, thus the results from both the logsum model and the travel time method can be compared to each other.

The travel time method measures the difference in travel time between the base scenario and any other scenario, and then multiplies that difference by the VOT for each trip purpose and the number of trips estimated for each trip purpose. For external trips, freight trips, and REC trips, these were all extracted directly from USTM. The HBW, HBO, and NHB purposes are estimated using the logsum model. Calculating the costs associated with the difference in travel time allows a better estimation of the true cost experienced by road users who are not included in the logsum based resiliency model. So, while the logit-based model is more sensitive to changes in the network, it is not able to truly estimate the costs associated with closure for those purposes that are not included in the analysis.

4.3. Comparative Scenario Results

We now apply the model to compare 40 additional scenarios where individual highway facilities are removed from the model highway network. These scenario locations are shown in Figure **??**, and were identified in a report by AEM (2017). AEM approached the idea of systemic resiliency by attempting to classify various types of threats toward specific infrastructure types. This method, while valid, was difficult to implement statewide. Some facilities have obvious natural threats such as earthquake, flood, or landslide.

This section will present the results of the 41 scenarios analyzed. First, Table **??** shows each of the scenarios, labeled "road10" for scenario 10, and "road11" for scenario 11 along with route numbers or street names, as well as a physical location where the link was cut.

The logsum method results are as follows in Table 14. The results are ranked from the road with the largest cost to the road with the smallest cost. We can see that road 50, which corresponds to I-80 between

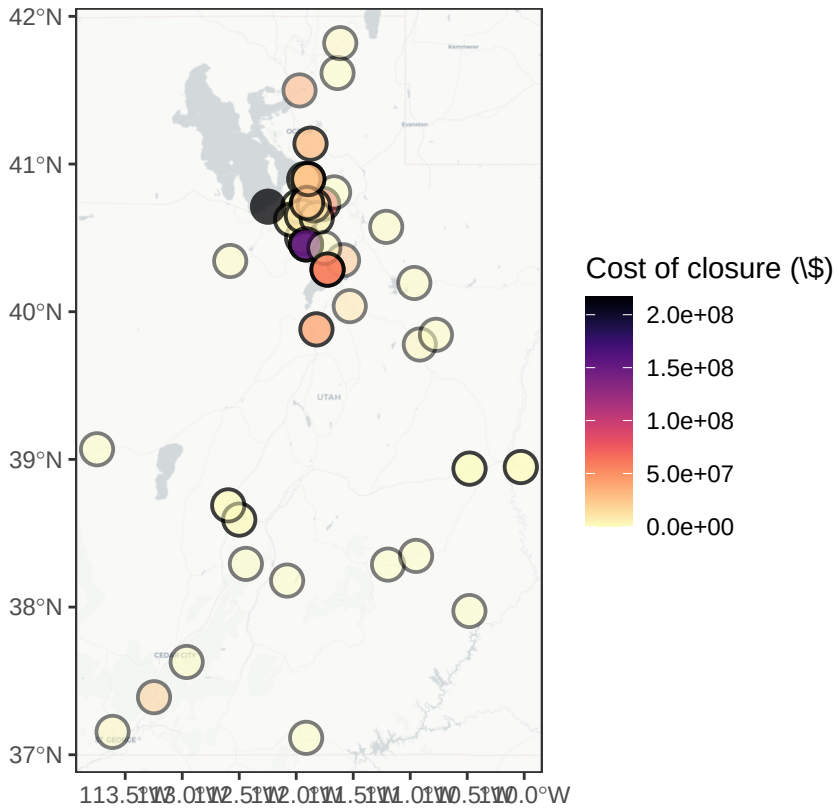


Figure 2: Total cost of link closure.

SLC and Tooele, experiences the largest cost per day according to the resiliency model. Following road 50, roads 38, 27, 37, and 14 make up the five most important road according to the cost estimation provided by the resiliency model. Each of these roads is an Interstate facility in Northern Utah, which is heavily populated. Some roads, such as road 10, which is along SR-95 near Hite, experiences no measurable change to HBW, HBO, or NHB traffic. This is likely due to the remote location of the highway link that was cut. A ranking is provided for all of the roads in Table **??**.

Scenario	Cost (per Day)	Route	Location
50	$2,170,329.71	I-80	between SLC and Tooele
38	$1,532,363.09	I-15	at the Point of the Mountain
37	$802,782.76	I-80	in Parley's Canyon
14	$574,474.74	I-15	in Orem between Univ. Ave & Center St
18	$411,179.06	I-15	near Rocky Ridge (between Payson & Nephi)
30	$370,133.75	US-91	between Brigham City & Mantua
27	$290,687.26	I-84	between Ogden and Morgan
12	$290,145.58	I-15	in Bountiful
46	$258,290.87	SR-189	up Provo Canyon near Vivian Park
40	$249,871.57	I-15	in SLC between 2100 S & 1300 S
20	$222,338.26	I-15	near New Harmony
39	$124,986.85	I-80	in SLC near Sugar House and 1300 E
41	$111,332.05	I-215	near Taylorsville
47	$105,106.54	US-6	up Spanish Fork Canyon near Diamond Fork Rd
43	$57,120.08	I-215	near Cottonwood Heights
34	$51,586.44	Bangerter	near Bluffdale
22	$32,372.99	US-6	in Carbon County North of Helper
17	$23,874.06	SR-18	just North of St. George
32	$21,432.72	US-89	between Logan and Bear Lake
42	$17,189.57	Bangerter	near West Valley City
44	$16,385.16	MVC (UT-85)	West of West Jordan
49	$10,756.48	I-70	near Richfield & Filmore
21	$9,246.15	US-40	East of Strawberry Reservoir
19	$8,071.17	I-15	near I-70 & Filmore
35	$7,562.58	SR-191	between Helper & Duchesne
15	$6,909.64	SR-199	near Rush Valley
24	$6,253.94	UT-35	outside of Francis
26	$5,239.14	SR-14	in Cedar Canyon
25	$4,286.78	Timp Highway	at the base of AF Canyon
48	$2,916.96	I-70	near Green River (NW of Moab)
45	$1,108.63	US-89	near the Border of Arizona by Lake Powell
23	$1,060.47	Legacy Parkway	near West Bountiful
31	$872.20	SR-62	East of Kingston
16	$362.54	SR-153	between Beaver & Junction
13	$357.00	I-70	at Dragon Point (W of Green River)
33	$275.22	SR-24	in Capitol Reef National Park
29	$226.44	SR-101	East of Hyrum
36	$223.53	SR-24	near Steamboat Point
11	$19.41	US-6	near King Top
28	$0.52	SR-65	on the border of Salt Lake County & Morgan County
10	$0.00	SR-95	near Hite

Table **??** contains the ranking results from the travel time method. Here, we see that the first five roads

differ from the results of the logsum model. Instead, road 48 becomes the most important road due to costs experienced. Road 48 is a part of I-70 near Green River, Utah. The other four roads that make up the top five most important roads in the travel time method analysis are road 13, road 20, road 50, and road 49. Some of these roads appear in both the logsum and travel time methods. Here again, several of the roads that are most important are located in Northern Utah. It is important to note that the main driving factor as to why a road was important or not in the travel time analysis was how much freight and external traffic it experienced. Including the freight, even with the logsum results, changes the rankings drastically.

Scenario	HBW, HBO, NHB	Freight, External, etc.	Total	HBW, HBO, NHB	Route	Location
48	$515.53	$1,603,380.58	$1,603,896.11	$2,916.96	I-70	near Green River (NW of Moab)
38	$827,013.58	$387,138.27	$1,214,151.85	$1,532,363.09	I-15	at the Point of the Mountain
50	$432,454.24	$419,624.71	$852,078.96	$2,170,329.71	I-80	between SLC and Tooele
20	$14,792.65	$646,271.16	$661,063.81	$222,338.26	I-15	near New Harmony
18	$49,815.00	$578,166.77	$627,981.77	$411,179.06	I-15	near Rocky Ridge (between Payson & Nephi)
19	$1,377.69	$393,559.50	$394,937.19	$8,071.17	I-15	near I-70 & Filmore
13	-$165.92	$320,400.54	$320,234.62	$357.00	I-70	at Dragon Point (W of Green River)
14	$137,033.68	$105,174.41	$242,208.10	$574,474.74	I-15	in Orem between Univ. Ave & Center St
37	$77,320.38	$164,493.06	$241,813.44	$802,782.76	I-80	in Parley's Canyon
27	$100,414.61	$87,561.49	$187,976.10	$290,687.26	I-84	between Ogden and Morgan
47	$2,212.81	$178,115.45	$180,328.26	$105,106.54	US-6	up Spanish Fork Canyon near Diamond Fork Rd
30	$55,514.96	$106,704.29	$162,219.25	$370,133.75	US-91	between Brigham City & Mantua
40	$98,707.14	$43,222.04	$141,929.18	$249,871.57	I-15	in SLC between 2100 S & 1300 S
49	-$13.32	$118,387.09	$118,373.77	$10,756.48	I-70	near Richfield & Filmore
12	$56,805.73	$53,575.26	$110,380.98	$290,145.58	I-15	in Bountiful
45	$314.60	$98,483.75	$98,798.34	$1,108.63	US-89	near the Border of Arizona by Lake Powell
39	$50,470.71	$41,831.37	$92,302.08	$124,986.85	I-80	in SLC near Sugar House and 1300 E
41	$79,619.29	$2,750.50	$82,369.79	$111,332.05	I-215	near Taylorsville
22	-$152.05	$62,555.97	$62,403.92	$32,372.99	US-6	in Carbon County North of Helper
46	$43,094.87	$13,098.33	$56,193.20	$258,290.87	SR-189	up Provo Canyon near Vivian Park
21	-$197.17	$56,200.18	$56,003.01	$9,246.15	US-40	East of Strawberry Reservoir
34	$42,503.10	$1,025.52	$43,528.62	$51,586.44	Bangerter	near Bluffdale
43	$34,850.96	$1,916.65	$36,767.61	$57,120.08	I-215	near Cottonwood Heights
42	$27,008.69	$503.00	$27,511.69	$17,189.57	Bangerter	near West Valley City
35	$31.23	$17,093.20	$17,124.43	$7,562.58	SR-191	between Helper & Duchesne
17	$12,131.00	$199.24	$12,330.24	$23,874.06	SR-18	just North of St. George
44	$10,854.46	$329.45	$11,183.92	$16,385.16	MVC (UT-85)	West of West Jordan
32	$1,244.65	$5,381.71	$6,626.37	$21,432.72	US-89	between Logan and Bear Lake

Scenario	HBW, HBO, NHB	Freight, External, etc.	Total	HBW, HBO, NHB	Route	Location
24	$2,845.57	$218.84	$3,064.40	$6,253.94	UT-35	outside of Francis
25	$2,566.09	$70.81	$2,636.90	$4,286.78	Timp Highway	at the base of AF Canyon
16	-$119.97	$2,665.35	$2,545.38	$362.54	SR-153	between Beaver & Junction
31	$778.55	$71.56	$850.11	$872.20	SR-62	East of Kingston
10	-$186.51	$858.65	$672.14	$0.00	SR-95	near Hite
26	$478.70	$136.01	$614.71	$5,239.14	SR-14	in Cedar Canyon
15	$335.46	$77.42	$412.88	$6,909.64	SR-199	near Rush Valley
23	$241.88	$42.07	$283.95	$1,060.47	Legacy Parkway	near West Bountiful
11	-$183.72	$423.05	$239.33	$19.41	US-6	near King Top
33	-$178.94	$328.21	$149.26	$275.22	SR-24	in Capitol Reef National Park
36	-$127.30	$227.21	$99.91	$223.53	SR-24	near Steamboat Point
29	-$2.72	$67.37	$64.65	$226.44	SR-101	East of Hyrum
28	-$186.37	$82.18	-$104.19	$0.52	SR-65	on the border of Salt Lake County & Morgan County

Some other interesting findings are that in the top 10 of each analysis methods, three scenarios appear in both rankings. Road 50, which corresponds to I-80 between Tooele and SLC, road 27 which corresponds to I-84 in Weber Canyon, and road 37 which corresponds to I-80 in Parley's Canyon are included in the top 10 scenarios for both methods of analysis. This is likely due to the number of passenger trips along these routes and the number of freight trips that occur along these routes.

4.4. Result Comparison: Logsum vs Travel Time

4.5. Positive Benefit Scenarios

Five of the scenarios indicated a benefit resulting from highway link closure. These scenarios were examined more closely to determine what causes these atypical results. The affected links are all located in Salt Lake Valley at the following locations: Bangerter Highway near Bluffdale, I-80 near 1300 E, I-15 between 2100 S and 1300 S, I-215 near Cottonwood Heights, and Mountain View Corridor near West Jordan.

It was determined that a likely cause which explains these atypical results is that the shortest path by time is not the shortest path by distance. The automobile accessibility is determined by the AM congested travel time in the Utah Statewide Travel Model (USTM). The travel distance – used to determine the accessibility of destinations by walking – is the distance of that path, and not the actual shortest distance path as might be more preferable.

When a highway link is broken, the new shortest path by time is longer than in the base scenario with this link available. But the new path may actually be shorter by distance. This causes an increase in the utility of accessing destinations by non-motorized modes, potentially overwhelming the decrease in automobile utility.

This is only observed in heavily urbanized regions for two reasons:

- The presence of high-speed expressways and parallel local roads means that alternate paths with shorter distances but longer vehicle times are more likely.
- The increased availability of destinations within the non-motorized distance threshold (50 miles) means that alternative destinations exist.

Overall, the results of the analysis indicate that the likely cause of a positive cost being estimated for these five scenarios is that there are easily accessible alternate routes in the area, along with competing TAZ of similar size in the DC size term equation.

4.6. Summary

The overall results show that the resiliency model is more sensitive to network changes than the travel time comparison. The ability for a user to choose both a mode and destination (or alternate destination) cause the logsum results to often estimate a smaller cost than the travel time results would. However, when the travel time results are factored in, the overall rankings of the 41 scenarios considered change dramatically. This is due to the large expenses experienced by freight traffic, which has a much higher VOT than other passenger trips do. In summary, Table **??** and Table **??** show the rankings for both the logsum and travel time analysis respectively. The logsum suggests that I-80 between Tooele and SLC is the most important road, while the travel time method, or total priority, indicates that I-70 near Green River is the most important road due to cost associated with closure.

5. Limitations and Discussion

Fixed transit travel time?
Shortest path: should be fixed?
Vary by purpose but not market segment
Travel time feedback diminished capacity
freight, external travel
how to do this with ABM?
sensitivity to utility parameters?

6. Conclusions

The USTM model is a gravity-based travel demand model, while the resiliency model is logit-based. The logit-based nature of the resiliency model allows for greater sensitivity in user mode and destination choice, which causes the estimated costs associated with link closure to be lower than for previous estimations. The resiliency models incorporation of both the logsum for HBW, HBW, and NHB purposes, as well as the travel time calculation for the other purposes included in USTM. The recommendations resulting from the adaptation of a logit-based model on the USTM network are that logit-based modeling returns more sensitive estimations of the value of a link in the network. This is a highly important outcome because more accurate estimation allows UDOT to better understand the monetary importance of highway links throughout Utah. Additionally, the model's design allows for further analysis of additional link closures, or even multi-link closure.

6.1. Summary

The development of a logit-based travel demand model can improve the ability of UDOT to accurately estimate the costs per day of link loss that Utahn's would experience. The resiliency model provides sensitive estimates that more accurately represent the costs associated with link closure than a travel time increase methodology by itself would be able to capture. Using the logsum, estimations sensitive to user choice were found, which can help professionals to better evaluate risk to Utah's infrastructure. User choice is highly important in modern modeling practices because user choice allows a model to estimate information more precisely and accurately. The information provided by the resiliency model should be used to prioritize link importance to the functionality of Utah's highway network. The resiliency model more accurately estimates costs experienced by Utahn's due to link loss than do traditional methods for determining costs.

Acknowledgments

This study was funded by the Utah Department of Transportation. The authors alone are responsible for the preparation and accuracy of the information, data, analysis, discussions, recommendations, and conclusions presented herein. The contents do not necessarily reflect the views, opinions, endorsements, or policies of the Utah Department of Transportation or the US Department of Transportation. The Utah Department of Transportation makes no representation or warranty of any kind, and assumes no liability therefore.

Author Contribution Statement

The authors confirm contribution to the paper as follows: study conception and design: G.S. Macfarlane; data collection: M. Barnes and N. Gray; analysis and interpretation of results: G.S. Macfarlane, M. Barnes, and N. Gray; draft manuscript preparation: G.S. Macfarlane, and M. Barnes. All authors reviewed the results and approved the final version of the manuscript.

References

Abdel-Rahim, A., Oman, P., Johnson, B., and Sadiq, R. (2007). Assessing surface transportation network component criticality: A multi-layer graph-based approach. In *2007 IEEE Intelligent Transportation Systems Conference*, pages 1000–1003.

AEM (2017). I-15 corridor risk and resilience pilot final report.

Agarwal, J., Liu, M., and Blockley, D. (2011). *A Systems Approach to Vulnerability Assessment*, pages 230–237.

Berdica, K. (2002). An introduction to road vulnerability: what has been done, is done and should be done. *Transport Policy*, 9(2):117–127.

Bradley, J. (2007). Time period and risk measures in the general risk equation. *Journal of Risk Research*, 10(3):355–369.

Dong, X., Ben-Akiva, M. E., Bowman, J. L., and Walker, J. L. (2006). Moving from trip-based to activity-based measures of accessibility. *Transportation Research Part A: Policy and Practice*, 40(2):163–180.

Geurs, K. T. and van Wee, B. (2004). Accessibility evaluation of land-use and transport strategies: review and research directions. *Journal of Transport Geography*, 12(2):127–140.

Hamedi, M., Eshragh, S., Franz, M., and Sekula, P. M. (2018). Analyzing impact of i-85 bridge collapse on regional travel in atlanta. Technical report.

Handy, S. L. and Niemeier, D. A. (1997). Measuring accessibility: An exploration of issues and alternatives. *Environment and Planning A: Economy and Space*, 29(7):1175–1194.

Ibrahim, S., Ammar, R., Rajasekaran, S., Lownes, N., Wang, Q., and Sharma, D. (2011). An efficient heuristic for estimating transportation network vulnerability. In *2011 IEEE Symposium on Computers and Communications (ISCC)*, pages 1092–1098.

Ip, W. H. and Wang, D. (2011). Resilience and friability of transportation networks: Evaluation, analysis and optimization. *IEEE Systems Journal*, 5(2):189–198.

Jaller, M., Calderón, C. A. G., Yushimito, W. F., and Díaz, I. D. S. (2015). An investigation of the effects of critical infrastructure on urban mobility in the city of medellín. *International Journal of Critical Infrastructures*, 11(3):213.

Koppelman, F. and Bhat, C. (2006). A self instructing course in mode choice modeling: Multinomial and nested logit models.

Omer, M., Mostashari, A., and Nilchiani, R. (2013). Assessing resilience in a regional road-based transportation network. *International Journal of Industrial and Systems Engineering*, 13(4):389–408.

Peeta, S., Sibel Salman, F., Gunnec, D., and Viswanath, K. (2010). Pre-disaster investment decisions for strengthening a highway network. *Computers & Operations Research*, 37(10):1708–1719.

Taylor, M. A. P. (2008). Critical transport infrastructure in urban areas: Impacts of traffic incidents assessed using accessibility-based network vulnerability analysis. *Growth and Change*, 39(4):593–616.

Vodák, R., Bíl, M., Svoboda, T., Křivánková, Z., Kubeček, J., Rebok, T., and Hliněný, P. (2019). A deterministic approach for rapid identification of the critical links in networks. *Plos One*, 14(7).

Xie, F. and Levinson, D. (2011). Evaluating the effects of the i-35w bridge collapse on road-users in the twin cities metropolitan region. *Transportation Planning and Technology*, 34(7):691–703.

Zhang, W. and Wang, N. (2016). Resilience-based risk mitigation for road networks. *Structural Safety*, 62:57–65.

Zhu, S., Levinson, D., Liu, H., Harder, K., and Dancyzk, A. (2010a). Traffic flow and road user impacts of the collapse of the i-35w bridge over the mississippi river.

Zhu, S., Levinson, D., Liu, H. X., and Harder, K. (2010b). The traffic and behavioral effects of the i-35w mississippi river bridge collapse. *Transportation Research Part A: Policy and Practice*, 44(10):771–784.